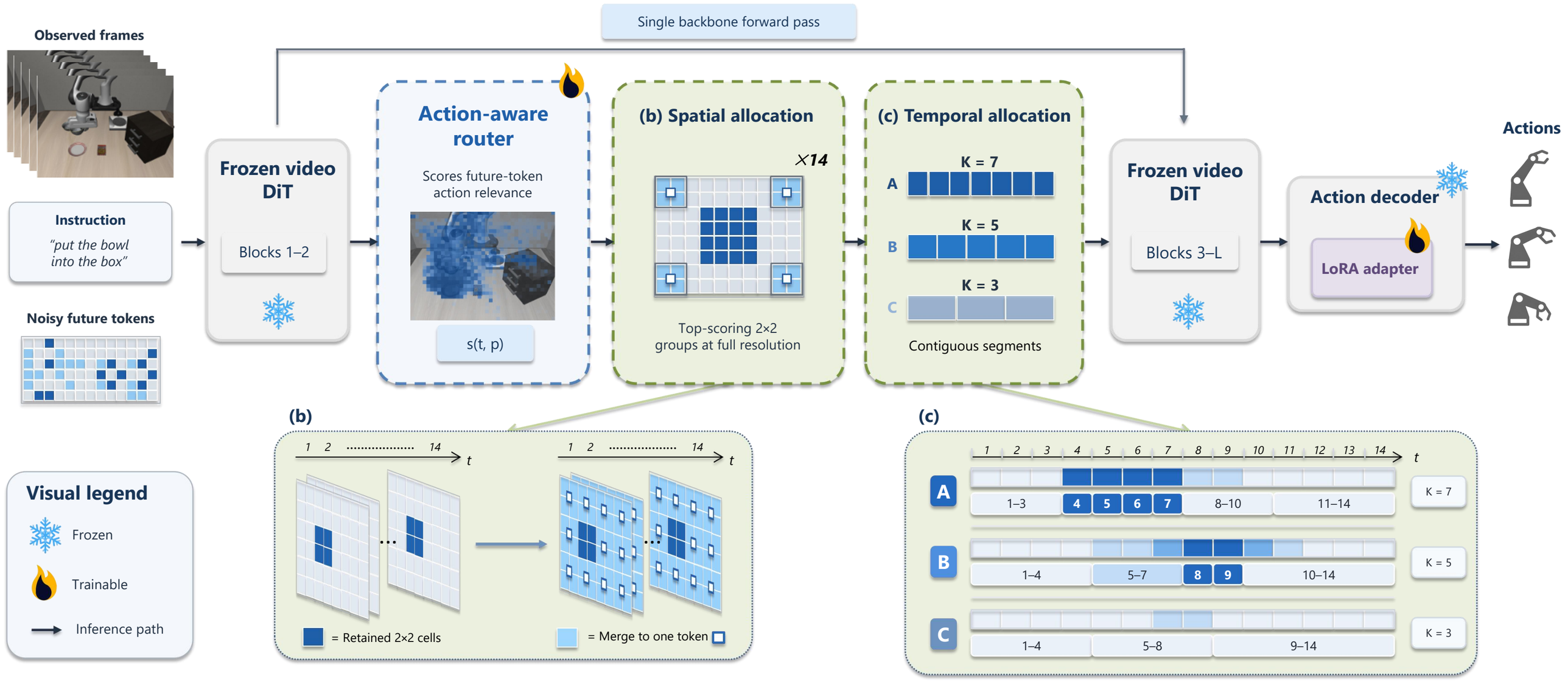

Single backbone forward pass
Observed frames
Instruction
"put the bowl into the box"
Noisy future tokens
Frozen video DiT
Blocks 1–2
Action-aware router
Scores future-token action relevance
s(t, p)
(b) Spatial allocation
×14
Top-scoring 2×2 groups at full resolution
(c) Temporal allocation
K = 7
A
K = 5
B
K = 3
C
Contiguous segments
Frozen video DiT
Blocks 3–L
Action decoder
LoRA adapter
Actions
Visual legend
Frozen
Trainable
Inference path
(b)
1 2 ..... 14 t
= Retained 2×2 cells
= Merge to one token
(c)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 t
A
1–3 4 5 6 7 8–10 11–14
K = 7
B
1–4 5–7 8 9 10–14
K = 5
C
1–4 5–8 9–14
K = 3